AI Capital versus Regular Capital: Output, Real Interest Rates, Wages and the Labor Share, and the Role of the Saving Model

Research note (draft)

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Abstract

How does technical progress in AI capital move output, the real interest rate, the real wage and the labor share, compared with progress in regular capital? Regular capital and labor form a Cobb-Douglas bundle, and AI capital substitutes for that bundle. Progress is either disembodied (in the production function) or embodied (a lower price of new capital in the accumulation equation). It arrives as a surprise jump, as an S-curve complete by year 10, or half and half. Every shock is a one-time level shift that raises long-run output by the same 10 percent. Under the standard Ramsey saving model, the long run is in closed form. AI progress raises the wage by $1/[1+(1-a)(\varepsilon-1)]$ of the output gain (74 percent here) and lowers the labor share, while regular-capital progress raises the wage one-for-one. In the short run, the real-rate response depends much more on whether progress is embodied and on its timing than on substitutability. Substitutability matters for r only after a surprise disembodied jump. That conclusion depends on the Ramsey assumption of perfectly elastic long-run capital supply. Under Solow (fixed saving), Diamond OLG, annual-frequency OLG and perpetual-youth saving, regular-capital progress still leaves long-run r unchanged. AI progress, however, changes long-run r exactly when $\varepsilon \neq 1$. For $\varepsilon = 1.5$, r rises by 1.1 percentage points under Solow, 0.4 under Diamond, 0.1 under the annual OLG and essentially zero under Ramsey.

1 Questions and main findings

The note asks three questions:

1. Do AI and regular-capital progress affect the interest rate, wages and the labor share differently, holding the long-run output gain fixed?
2. How much of the difference comes from AI's substitutability for labor and capital?
3. How much depends on the saving model, given that the Ramsey Euler equation makes long-run capital supply perfectly elastic?

Main findings:

- *Wages and the labor share* differ sharply between AI and regular-capital progress. The difference comes from substitutability, and it holds in every saving model.

- *The real interest rate* under Ramsey depends mainly on whether progress is embodied and on its timing. Substitutability matters little, except after surprise disembodied jumps.
- *Long-run interest rates* respond to AI progress, and through substitutability, once saving is not perfectly elastic. Regular-capital progress inside the Cobb-Douglas bundle never moves long-run r.

2 Model

Notation follows Kiley (2000). Production is

$$X = (\Gamma_K K)^a (ZN)^{1-a}, \qquad Y = \left[\omega^{1/\varepsilon} X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} (\Gamma_A A)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon = 1.5 \text{ (baseline)}.$$

Capital $j \in \{K, A\}$ is measured in efficiency units: $K_j' = (1-d_j)K_j + I_j/P_j$. *Disembodied* progress raises Γ_j, which improves installed capital. *Embodied* progress lowers P_j, which improves only new investment. Hours are fixed and trend growth is $g = 1.5$ percent.

Experiments. Each is a one-time, permanent level shift in Γ_j or $1/P_j$, announced at $t = 0$, with three timings:

- a *jump* at $t = 0$;
- an *S-curve*, $S_t = \frac{1}{2}[1 - \cos(\pi t/10)]$, which starts at $t = 0$ and is complete by year 10;
- *half/half*: half the log improvement as a jump, half along the S-curve.

Elasticities are the same in every experiment. Each shock is scaled *within each model* so that long-run output, relative to the pre-shock path, rises by 10 percent.

Calibration. $a = 0.282$, $d_K = 0.066$, initial $r = 7.5$ percent, initial AI income share 3 percent, initial labor share 69.6 percent. Two accumulation settings are used:

(A) $d_A = 0.20$, with the adjustment costs of Kiley (2004), $P_j \frac{c_j}{2}(i_j - x_j)^2 K_j$, where $i_j = I_j/(P_j K_j)$, $x_j = g + d_j$ and $c_j = 1/x_j$, so that both stocks have a unit elasticity of investment to Q;

(B) $d_A = d_K = 0.066$ and no adjustment costs. This isolates substitutability from differences in depreciation and adjustment costs.

3 Closed-form long run

Proposition 1 (Balanced growth path)**.** *With constant technology levels and real rate r, $Q_j = P_j$ and $u_j = r + d_j$. With $\lambda_j = P_j u_j/\Gamma_j$,*

$$s_A = (1-\omega)\lambda_A^{1-\varepsilon}, \quad \textit{labor share} = (1-a)(1-s_A), \quad P_X = \left(\frac{1-s_A}{\omega}\right)^{\frac{1}{1-\varepsilon}},$$
$$\ln\frac{W}{Z} = \frac{\ln P_X}{1-a} - \frac{a}{1-a}\ln\lambda_K + \textit{const}, \qquad \ln\frac{Y}{Z} = \ln\frac{W}{Z} - \ln(1-s_A) + \textit{const}.$$

Proposition 2 (Same output gain γ_Y, given r)**.** *Disembodied and embodied progress have the same long-run effects* ($\ln \Gamma_j = -\ln P_j$)*.* Regular capital*:* $\ln(\Gamma_K/P_K) = \frac{1-a}{a}\gamma_Y$, $\Delta \ln W = \gamma_Y$, *and the labor share is unchanged.* AI*:* $\Delta \ln W = \dfrac{\gamma_Y}{1+(1-a)(\varepsilon-1)}$ *and* $\Delta \ln(\textit{labor share}) = -\dfrac{(1-a)(\varepsilon-1)}{1+(1-a)(\varepsilon-1)}\gamma_Y$.

Under Ramsey, r is pinned at $(1+g)/B - 1$, so Proposition 2 gives the long run directly. For $\gamma_Y = 10$ percent, regular-capital technology rises 29 percent and AI technology 3.4-fold. The wage rises 10.0 percent (regular capital) or 7.4 percent (AI), and under AI progress the labor share falls from 69.6 to 67.8 percent.

4 Dynamics under Ramsey

4.1 Setting A: AI depreciates faster and both stocks face adjustment costs

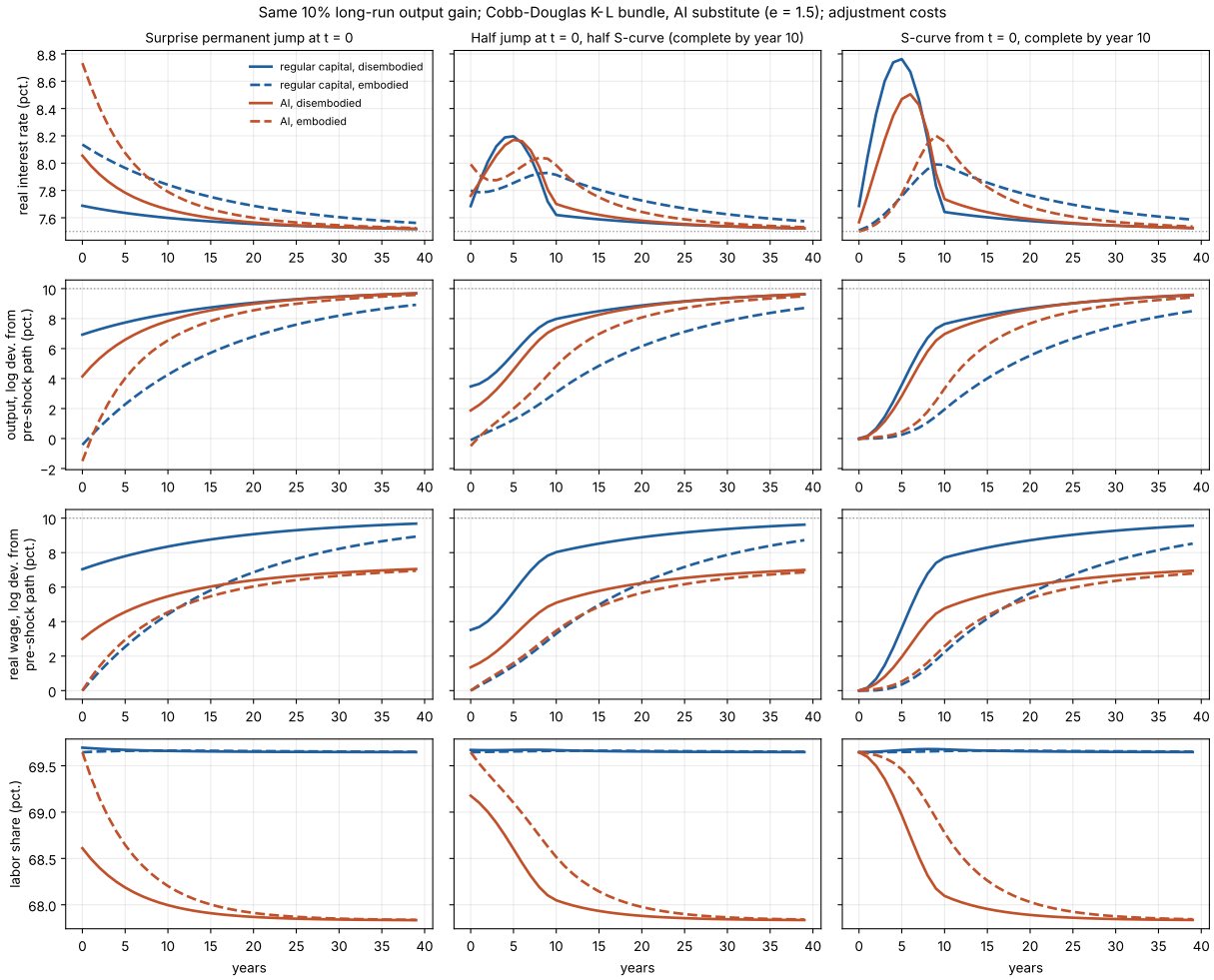


Figure 1: Setting A ($d_A = 0.20$, adjustment costs). Same 10% long-run output gain.

Table 1: Setting A. r_1 is the rate between years 0 and 1. Output and wages are log deviations (percent) from the pre-shock path.

		real rate (pct.)			output, $\ln Y$ (pct.)			real wage, $\ln W$ (pct.)			labor share
		r_1	peak	year	t=0	t=10	t=20	t=0	t=10	t=20	t=10 (pct.)
Jump	Regular, disemb.	7.69	7.69	0	6.9	8.3	9.1	7.0	8.4	9.1	69.7
	Regular, emb.	8.14	8.14	0	-0.4	4.3	6.8	0.0	4.4	6.9	69.7
	AI, disemb.	8.05	8.05	0	4.1	7.8	9.0	3.0	5.5	6.4	68.0
	AI, emb.	8.73	8.73	0	-1.5	6.6	8.6	0.0	4.5	6.0	68.2
Half/half	Regular, disemb.	7.69	8.20	5	3.5	8.0	8.9	3.5	8.0	8.9	69.7
	Regular, emb.	7.80	7.93	9	-0.1	3.1	6.2	0.0	3.3	6.2	69.7
	AI, disemb.	7.76	8.17	5	1.9	7.4	8.8	1.4	5.1	6.2	68.0
	AI, emb.	7.99	8.04	8	-0.5	4.8	8.1	0.0	3.5	5.7	68.5
S-curve	Regular, disemb.	7.69	8.76	5	0.0	7.6	8.7	0.0	7.7	8.7	69.7
	Regular, emb.	7.51	7.99	9	0.0	1.9	5.5	0.0	2.2	5.6	69.7
	AI, disemb.	7.57	8.50	6	-0.1	7.0	8.6	0.0	4.8	6.1	68.1
	AI, emb.	7.50	8.20	9	0.0	3.3	7.7	0.0	2.6	5.4	68.8

- *Real rate.* After a jump, r rises most for embodied AI (8.7 percent), then embodied regular capital (8.1), disembodied AI (8.05) and disembodied regular capital (7.7). Along the S-curve it peaks earlier and higher under disembodied progress: 8.8 (regular) and 8.5 (AI) around years 5–6, against 8.0 and 8.2 around year 9 under embodied progress. The half/half case lies between the two.

- *Output.* Disembodied progress raises output on impact (6.9 percent regular, 4.1 AI). An embodied jump briefly *lowers* net output (by 0.4 and 1.5 percent), because the investment boom absorbs resources in adjustment costs. AI output catches up quickly because AI capital turns over fast.

- *Wages and the labor share.* AI wages lag AI output, because the AI share rises. The labor share moves only under AI progress.

4.2 Setting B: equal depreciation and no adjustment costs

Table 2: Setting B. Columns as in Table 1.

		real rate (pct.)			output, $\ln Y$ (pct.)			real wage, $\ln W$ (pct.)			labor share
		r_1	peak	year	t=0	t=10	t=20	t=0	t=10	t=20	t=10 (pct.)
Jump	Regular, disemb.	8.33	8.33	0	7.0	9.4	9.9	7.0	9.4	9.9	69.7
	Regular, emb.	10.38	10.38	0	0.0	8.1	9.6	0.0	8.1	9.6	69.7
	AI, disemb.	8.80	8.80	0	4.5	8.8	9.7	3.0	6.2	7.1	67.9
	AI, emb.	9.51	9.51	0	0.0	8.3	9.6	0.0	5.7	7.0	67.9
Half/half	Regular, disemb.	7.95	8.05	6	3.5	8.6	9.7	3.5	8.7	9.7	69.7
	Regular, emb.	8.72	8.72	0	0.0	5.7	9.1	0.0	5.8	9.1	69.7
	AI, disemb.	8.06	8.17	7	2.0	8.0	9.5	1.4	5.5	6.9	67.9
	AI, emb.	8.29	8.53	10	0.0	5.9	9.0	0.0	3.7	6.5	68.1
S-curve	Regular, disemb.	7.57	8.30	7	0.0	7.9	9.6	0.0	8.0	9.6	69.7
	Regular, emb.	7.14	9.36	10	0.0	3.4	8.7	0.0	3.6	8.7	69.8
	AI, disemb.	7.51	8.35	8	0.0	7.3	9.4	0.0	4.8	6.8	67.9
	AI, emb.	7.24	8.83	10	0.0	4.6	8.8	0.0	2.7	6.2	68.3

Without adjustment costs, investment responds faster and r moves more.

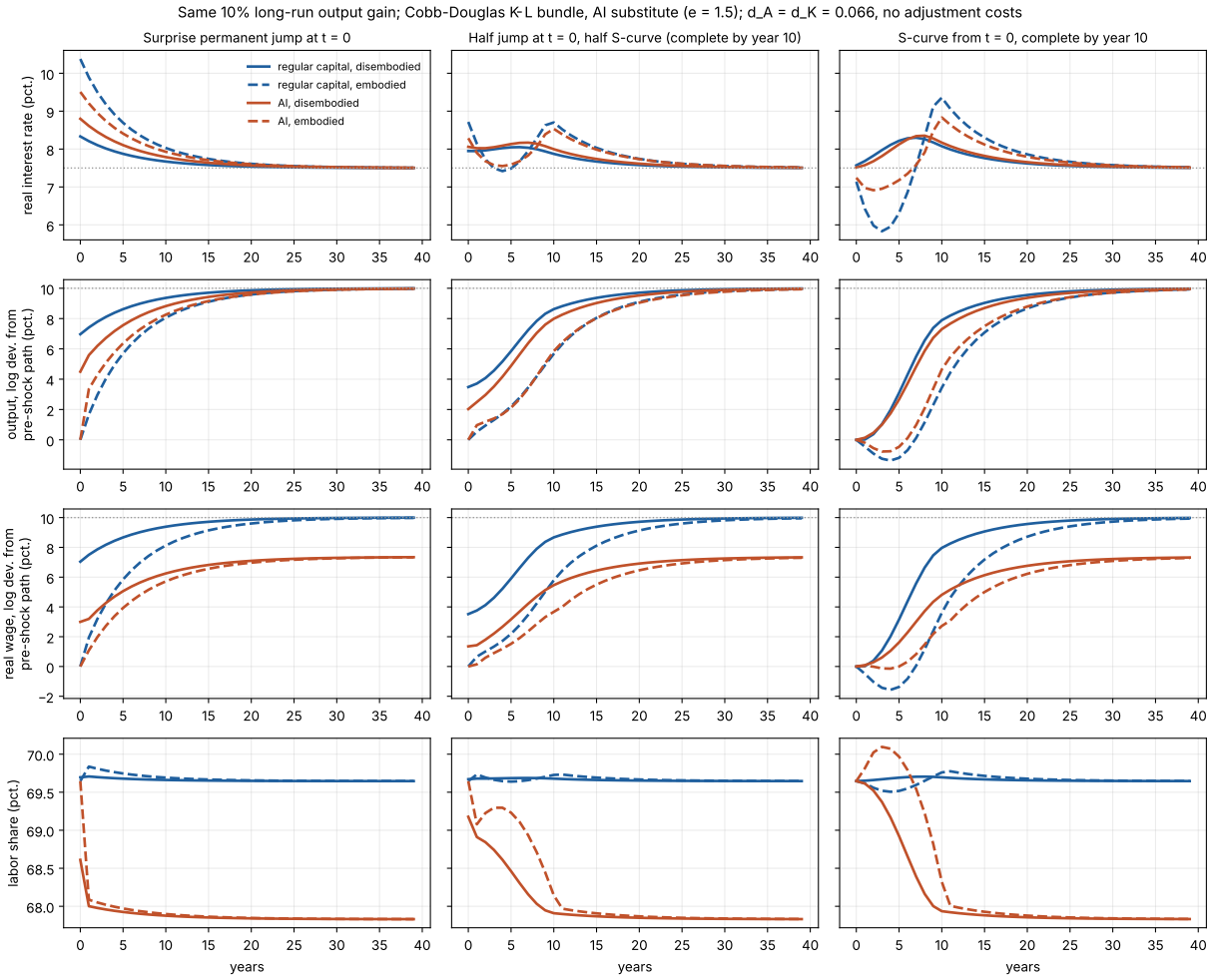


Figure 2: Setting B ($d_A = d_K = 0.066$, no adjustment costs).

- *Embodied regular-capital progress* has the largest impact effect, 10.4 percent after a jump. Regular capital is a much larger stock than AI, so it requires the largest investment swing.
- *An anticipated embodied S-curve* produces a two-phase response. Firms wait for cheaper capital, so r first *falls* (to 5.8 percent for regular capital and 6.9 for AI) and output and wages dip. Then an investment boom pushes r to 9.4 (regular) or 8.8 percent (AI) at year 10.
- *Capital adjusts within a year*, so after an AI jump the labor share falls almost entirely by year 1.

4.3 How much does substitutability matter for r?

Table 3 varies only AI's elasticity in Setting B, from 0.75 (near the complements limit, where a 10 percent output gain needs a roughly 150-fold improvement in AI technology) to 2. Comparing AI and regular capital directly confounds substitutability with AI's small initial share. This experiment does not.

Table 3: Ramsey, Setting B, AI progress, varying ε. "40-yr excess" is $\sum_{t<40}[\ln(1+r_{t+1}) - \ln 1.075]$ in percentage points.

	jump, disemb.		jump, emb.		S-curve, disemb.	S-curve, emb.	long run	
ε (AI)	r_1	40-yr excess	r_1	40-yr excess	peak	trough/peak	$\Delta \ln W$	labor share
0.75	7.96	2.8	9.10	9.3	8.26	6.74/8.54	12.2	71.2
1	8.36	5.4	9.38	11.3	8.31	6.90/8.79	10.0	69.6
1.5	8.80	8.6	9.51	13.0	8.35	6.91/8.83	7.4	67.8
2	8.95	10.2	9.46	13.5	8.36	6.91/8.78	5.8	66.8

- *Along an S-curve and for embodied shocks*, ε barely moves r. Peaks and troughs change by 0.1–0.4 points across the whole range.
- *After a surprise disembodied jump*, ε matters. r_1 rises from 8.0 to 9.0 percent, and the 40-year cumulative excess return from 2.8 to 10.2 points. With higher ε, less of the output gain arrives on impact, more must come from accumulation, and expected consumption growth is higher.
- *Wages and the labor share* are highly sensitive to ε. The long-run wage gain falls from 12.2 to 5.8 percent, and the labor share from 71.2 to 66.8 percent.

At $\varepsilon = 1$, disembodied AI and disembodied regular-capital progress produce identical paths, as theory implies.

5 The saving model

Why Ramsey pins r. With an infinitely lived household, the steady-state Euler equation fixes $1 + r = (1 + g)/B$, so long-run capital supply is perfectly elastic. Saving adjusts to whatever is needed to deliver that rate, and no level shift in technology can move long-run r. The closed-form CES-pairing models of the companion note have constant saving *rates*, but they too pin r by B and g.

Alternatives (all with Setting B, $\varepsilon = 1.5$ unless noted, and recalibrated to the same initial path):

- *Solow*: a fixed gross saving rate s.
- *Perpetual youth* (Blanchard, 1985; Yaari, 1965): death probability $p = 0.025$ (a 40-year horizon), annuities, log utility and flat labor income, so that $C = [1 - B(1 - p)](A + H)$, where H is human wealth discounted at $(1 + r)/(1 - p)$. With $p = 0$ this is Ramsey.
- *Annual OLG*: 60 cohorts (ages 20–79) who work for 45 years with flat earnings, log utility and no bequests. Cohorts alive at $t = 0$ hold their steady-state wealth, revalued by the shock.
- *Diamond OLG* (Diamond, 1965): two 30-year periods, with the young saving $B_L/(1 + B_L)$ of wages. Long run only.

Proposition 3 (Long-run interest rate)**.** *In every model, the capital share is* $1-(1-a)(1-s_A)$ *and wealth relative to output is (capital share)*$/(r+d)$*. Long-run* r *solves:*

$$\begin{aligned} \textit{Ramsey:} \quad & 1+r=(1+g)/B; \\ \textit{Solow:} \quad & (g+d)\,\frac{\textit{capital share}}{r+d}=s; \\ \textit{Diamond:} \quad & r_L+d_L=(1+g)^L\,\frac{\textit{capital share}}{s_y\cdot\textit{labor share}}; \\ \textit{Annual OLG, perpetual youth:} \quad & \textit{household wealth}(r)\cdot\textit{labor share}=(1+r)\,\frac{\textit{capital share}}{r+d}, \end{aligned}$$

with household wealth per unit of labor income in closed form for each (a 60-term sum for the OLG; a geometric sum for perpetual youth). With a Cobb-Douglas bundle, regular-capital progress leaves the capital share unchanged at every r*, so it never changes long-run* r*. AI progress changes the capital share if and only if* $\varepsilon\neq 1$*. It then raises* r *if* $\varepsilon>1$ *and lowers it if* $\varepsilon<1$*, in every model where capital supply is not perfectly elastic.*

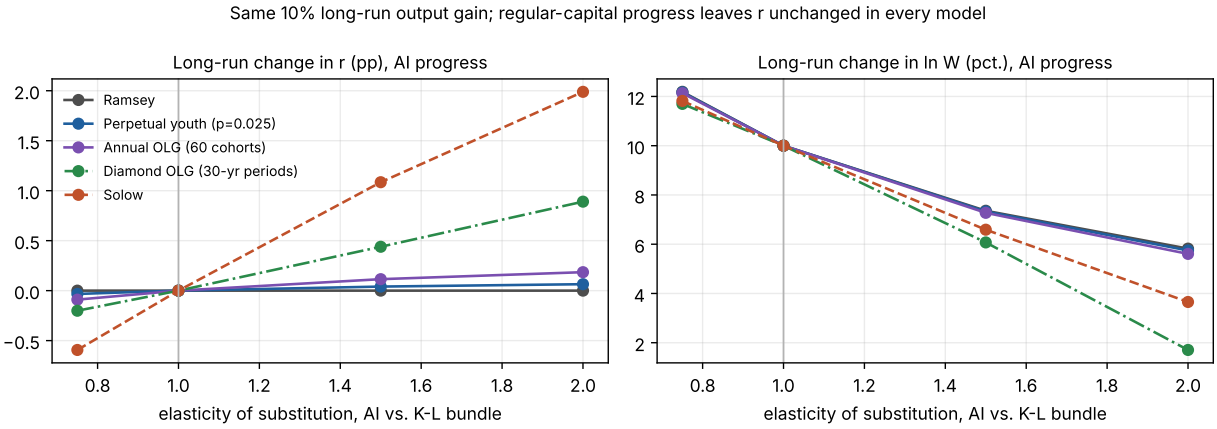


Figure 3: Long-run effects of AI progress by saving model and elasticity (same 10% output gain).

Long-run results (Table 4, Figure 3).

- *Regular-capital progress* leaves r unchanged in every model, and wages rise one-for-one with output.
- *AI progress with* $\varepsilon=1.5$ raises r by 0 (Ramsey), 0.04 (perpetual youth), 0.11 (annual OLG), 0.44 (Diamond) and 1.08 points (Solow). At $\varepsilon=2$ the figures are 0, 0.06, 0.18, 0.89 and 1.99. At $\varepsilon=0.75$, r falls in every model except Ramsey.
- *The less elastic the saving, the more the AI shock must deliver and the less labor gets.* With the same 10 percent output gain, the required AI improvement rises from 3.4-fold (Ramsey) to 4.7-fold (Solow) and 5.7-fold (Diamond). The wage gain falls from 7.4 to 6.6 and 6.1 percent.
- *Why the two OLG models differ.* Life-cycle saving in the annual OLG is very elastic to r: $B(1+r)/(1+g)\approx 0.995$, so a small rise in r calls forth much more saving. Its long-run responses are therefore close to Ramsey. In the Diamond model, saving out of wages makes the

Table 4: Long run by saving model. The shock is the log improvement in AI (or regular-capital) technology needed for a 10% output gain.

ε	saving model	shock (log)	Δr (pp)	long-run r	$\Delta \ln W$	labor share	AI share
0.75	Ramsey	5.02	+0.00	7.50	12.2	71.2	0.9
0.75	Perpetual youth (p=0.025)	4.93	-0.03	7.47	12.2	71.2	0.9
0.75	Annual OLG (60 cohorts)	4.77	-0.09	7.41	12.1	71.1	0.9
0.75	Diamond OLG (30-yr periods)	3.17	-0.20	7.30	11.7	70.8	1.3
0.75	Solow	3.56	-0.59	6.91	11.8	70.9	1.2
1	Ramsey	2.32	+0.00	7.50	10.0	69.6	3.0
1	Perpetual youth (p=0.025)	2.32	+0.00	7.50	10.0	69.6	3.0
1	Annual OLG (60 cohorts)	2.32	+0.00	7.50	10.0	69.6	3.0
1	Diamond OLG (30-yr periods)	2.32	+0.00	7.50	10.0	69.6	3.0
1	Solow	2.32	+0.00	7.50	10.0	69.6	3.0
1.5	Ramsey	1.22	+0.00	7.50	7.4	67.8	5.5
1.5	Perpetual youth (p=0.025)	1.24	+0.04	7.54	7.3	67.8	5.6
1.5	Annual OLG (60 cohorts)	1.26	+0.11	7.61	7.3	67.8	5.6
1.5	Diamond OLG (30-yr periods)	1.74	+0.44	7.94	6.1	67.0	6.7
1.5	Solow	1.54	+1.08	8.58	6.6	67.3	6.3
2	Ramsey	0.84	+0.00	7.50	5.8	66.8	7.0
2	Perpetual youth (p=0.025)	0.86	+0.06	7.56	5.7	66.7	7.0
2	Annual OLG (60 cohorts)	0.88	+0.18	7.68	5.6	66.7	7.2
2	Diamond OLG (30-yr periods)	1.52	+0.89	8.39	1.7	64.1	10.7
2	Solow	1.23	+1.99	9.49	3.7	65.4	9.0
reg. capital	Ramsey	0.25	+0.00	7.50	10.0	69.6	3.0
	Perpetual youth (p=0.025)	0.25	+0.00	7.50	10.0	69.6	3.0
	Annual OLG (60 cohorts)	0.25	+0.00	7.50	10.0	69.6	3.0
	Diamond OLG (30-yr periods)	0.25	+0.00	7.50	10.0	69.6	3.0
	Solow	0.25	+0.00	7.50	10.0	69.6	3.0

proportional change in the 30-year return larger, but compounding over the period translates it into a moderate change in the annual rate. Perpetual youth with flat labor income and a 40-year horizon stays close to Ramsey.

Transitions (Figure 4; full table in the appendix).

- *Perpetual youth and the annual OLG track Ramsey closely* (within about 0.1 point for r), apart from the small permanent rise in r after AI progress.
- *Solow amplifies and prolongs the AI response, and makes it permanent.* After a jump, r_1 is 9.5 percent (disembodied) or 10.4 percent (embodied), against 8.8 and 9.5 under Ramsey. Along the S-curve it peaks at 9.4 (disembodied) or 10.0 (embodied), against 8.35 and 8.8. Then r converges to 8.6 rather than 7.5 percent. Wages rise more slowly.
- *For regular-capital progress* the saving model matters only in the transition. Solow gives larger embodied swings (S-curve trough 5.4 and peak 9.9 percent), and r returns to 7.5 percent.
- *The embodied-anticipation dip in r* appears under every saving model.

6 Summary

1. With the same long-run output gain, AI progress raises wages by about three-quarters as much as regular-capital progress and lowers the labor share. This holds in closed form, for

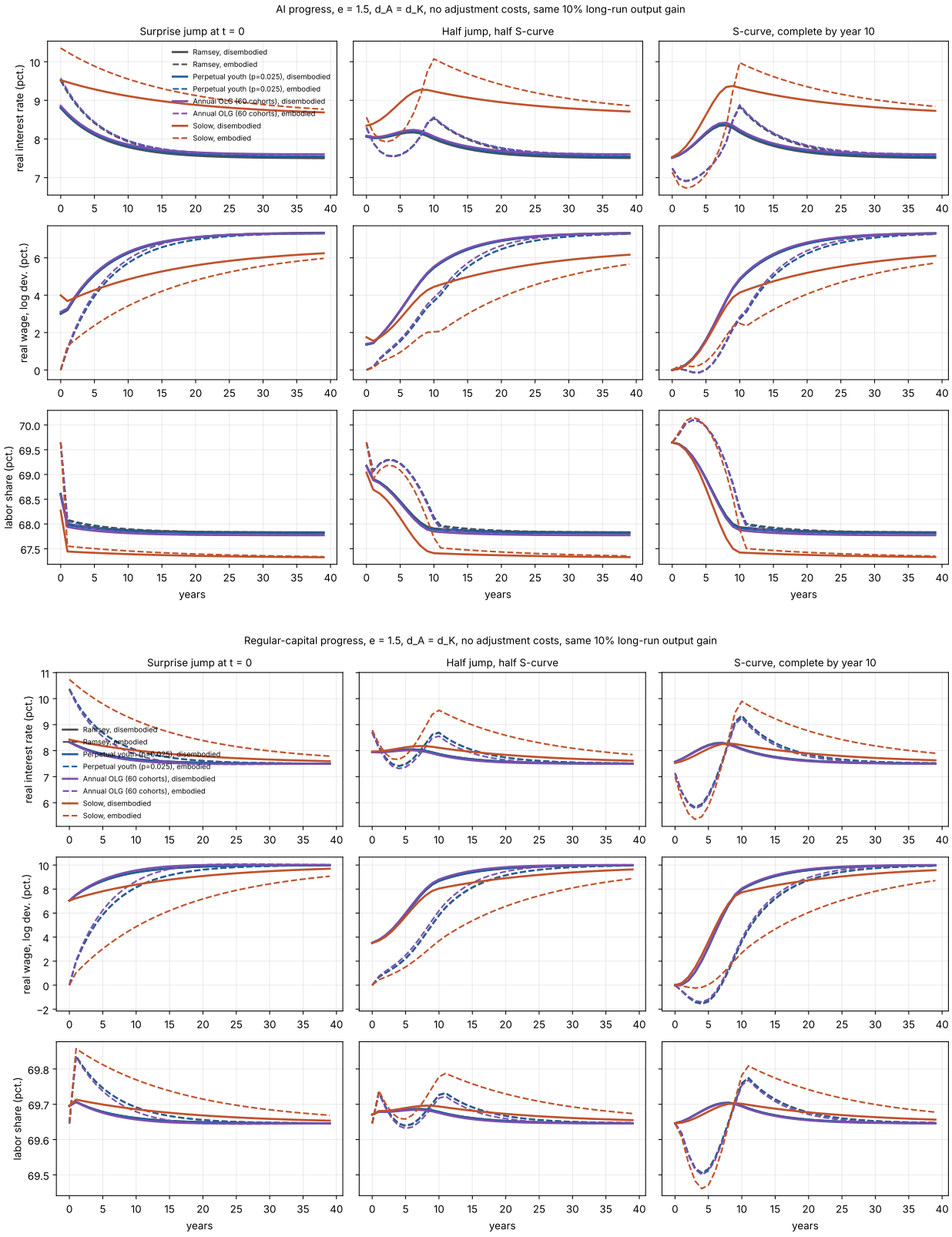


Figure 4: Transitions by saving model, Setting B, $\varepsilon = 1.5$. Top: AI progress. Bottom: regular-capital progress.

embodied and disembodied progress alike, and in every saving model. Less elastic saving widens the gap.

2. Under Ramsey, the real-rate response depends mainly on whether progress is embodied and on its timing: jumps versus anticipated S-curves, and the wait-for-cheaper-capital dip. Substitutability matters only after surprise disembodied jumps.

3. The Ramsey result that substitutability barely moves r comes from perfectly elastic long-run capital supply. With fixed saving (Solow) or saving out of wages (Diamond), AI progress with $\varepsilon > 1$ raises long-run r permanently, by up to 2 points at $\varepsilon = 2$. Regular-capital progress does not. Annual life-cycle (OLG) and perpetual-youth saving are elastic enough that the effect is small, 0.1–0.2 points.

A Transitions by saving model

Table 5: Transitions by saving model, Setting B, $\varepsilon = 1.5$. "Long run" r is the closed form of Proposition 3.

			real rate (pct.)			$\ln W$ (pct.)		labor share
timing	progress	saving model	r_1	max	long run	t=10	long run	t=10
Jump	AI disemb.	Ramsey	8.80	8.80	7.50	6.2	7.4	67.9
		Perpetual youth	8.82	8.82	7.54	6.3	7.3	67.9
		Annual OLG	8.86	8.86	7.61	6.3	7.3	67.8
		Solow	9.52	9.52	8.58	4.8	6.6	67.4
	AI emb.	Ramsey	9.51	9.51	7.50	5.7	7.4	67.9
		Perpetual youth	9.53	9.53	7.54	5.7	7.3	67.9
		Annual OLG	9.57	9.57	7.61	5.9	7.3	67.8
		Solow	10.35	10.35	8.58	3.4	6.6	67.5
	Reg. disemb.	Ramsey	8.33	8.33	7.50	9.4	10.0	69.7
		Perpetual youth	8.33	8.33	7.50	9.4	10.0	69.7
		Annual OLG	8.32	8.32	7.50	9.6	10.0	69.7
		Solow	8.42	8.42	7.50	8.4	10.0	69.7
	Reg. emb.	Ramsey	10.38	10.38	7.50	8.1	10.0	69.7
		Perpetual youth	10.37	10.37	7.50	8.2	10.0	69.7
		Annual OLG	10.33	10.33	7.50	8.7	10.0	69.7
		Solow	10.74	10.74	7.50	4.9	10.0	69.8
Half/half	AI disemb.	Ramsey	8.06	8.17	7.50	5.5	7.4	67.9
		Perpetual youth	8.07	8.20	7.54	5.5	7.3	67.9
		Annual OLG	8.08	8.24	7.61	5.5	7.3	67.8
		Solow	8.34	9.28	8.58	4.4	6.6	67.4
	AI emb.	Ramsey	8.29	8.53	7.50	3.7	7.4	68.1
		Perpetual youth	8.29	8.56	7.54	3.7	7.3	68.1
		Annual OLG	8.30	8.56	7.61	3.9	7.3	68.1
		Solow	8.56	10.07	8.58	2.0	6.6	67.7
	Reg. disemb.	Ramsey	7.95	8.05	7.50	8.7	10.0	69.7
		Perpetual youth	7.95	8.04	7.50	8.7	10.0	69.7
		Annual OLG	7.94	8.01	7.50	8.8	10.0	69.7
		Solow	7.97	8.18	7.50	8.0	10.0	69.7
	Reg. emb.	Ramsey	8.72	8.72	7.50	5.8	10.0	69.7
		Perpetual youth	8.71	8.71	7.50	5.9	10.0	69.7
		Annual OLG	8.69	8.69	7.50	6.2	10.0	69.7
		Solow	8.79	9.55	7.50	3.7	10.0	69.8
S-curve	AI disemb.	Ramsey	7.51	8.35	7.50	4.8	7.4	67.9
		Perpetual youth	7.51	8.38	7.54	4.8	7.3	67.9
		Annual OLG	7.52	8.42	7.61	4.9	7.3	67.9
		Solow	7.53	9.37	8.58	4.1	6.6	67.4
	AI emb.	Ramsey	7.24	8.83	7.50	2.7	7.4	68.3
		Perpetual youth	7.24	8.86	7.54	2.7	7.3	68.3
		Annual OLG	7.23	8.88	7.61	2.8	7.3	68.3
		Solow	7.14	9.97	8.58	2.5	6.6	67.9
	Reg. disemb.	Ramsey	7.57	8.30	7.50	8.0	10.0	69.7
		Perpetual youth	7.57	8.29	7.50	8.0	10.0	69.7
		Annual OLG	7.56	8.27	7.50	8.1	10.0	69.7
		Solow	7.52	8.28	7.50	7.7	10.0	69.7
	Reg. emb.	Ramsey	7.14	9.36	7.50	3.6	10.0	69.8
		Perpetual youth	7.13	9.33	7.50	3.7	10.0	69.8
		Annual OLG	7.12	9.26	7.50	3.9	10.0	69.8
		Solow	7.00	9.90	7.50	2.7	10.0	69.8

B Computation and verification

All results are reproduced by `capital_cases.py` (Settings A and B, run with the `--equal-d-noadj` switch) and `saving_models.py` (saving models and the elasticity grid). Each transition is a non-linear perfect-foresight solve of the full equilibrium.

- Ramsey, Solow and perpetual youth use Newton's method on the capital paths, with residuals below 10^{-10}.
- The annual OLG uses Levenberg-Marquardt started from the Ramsey solution. Plain Newton overshoots because life-cycle saving is very sensitive to returns. Residuals are below 10^{-9}. The OLG is solved over 300 years and its long run is read at year 200, because the final cohorts carry a horizon-truncation artifact.

Further checks:

- Perpetual youth with $p = 0$ reproduces the Euler-equation Ramsey solver.
- Steady-state OLG household wealth equals the value of capital.
- Every experiment converges to the 10 percent output gain and to the closed-form long-run r.
- Long-run wages match Proposition 2.

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Table 6: Verification, saving models (largest errors).

Check	max abs. error
OLG: household assets = (1+r)V on initial BGP	1.9e-13
OLG: lifetime budget (terminal assets)	1.2e-14
ramsey e=0.75: initial r reproduces 7.5%	0.0e+00
py e=0.75: initial r reproduces 7.5%	6.9e-17
olg e=0.75: initial r reproduces 7.5%	1.5e-16
diamond e=0.75: initial r reproduces 7.5%	4.2e-17
solow e=0.75: initial r reproduces 7.5%	1.4e-17
ramsey e=1.0: initial r reproduces 7.5%	0.0e+00
py e=1.0: initial r reproduces 7.5%	6.9e-17
olg e=1.0: initial r reproduces 7.5%	1.5e-16
diamond e=1.0: initial r reproduces 7.5%	4.2e-17
solow e=1.0: initial r reproduces 7.5%	1.4e-17
ramsey e=1.5: initial r reproduces 7.5%	0.0e+00
py e=1.5: initial r reproduces 7.5%	6.9e-17
olg e=1.5: initial r reproduces 7.5%	1.5e-16
diamond e=1.5: initial r reproduces 7.5%	4.2e-17
solow e=1.5: initial r reproduces 7.5%	1.4e-17
ramsey e=2.0: initial r reproduces 7.5%	0.0e+00
py e=2.0: initial r reproduces 7.5%	6.9e-17
olg e=2.0: initial r reproduces 7.5%	1.5e-16
diamond e=2.0: initial r reproduces 7.5%	4.2e-17
solow e=2.0: initial r reproduces 7.5%	1.4e-17
ramsey jump K dis: residual	2.4e-14
ramsey jump K dis: long-run ln Y = 10%	9.3e-16
ramsey jump K dis: long-run r = closed form	4.2e-17
ramsey jump K emb: residual	2.3e-15
ramsey jump K emb: long-run ln Y = 10%	7.4e-16
ramsey jump K emb: long-run r = closed form	4.2e-17
ramsey jump A dis: residual	2.7e-13
ramsey jump A dis: long-run ln Y = 10%	7.4e-16
ramsey jump A dis: long-run r = closed form	4.2e-17
ramsey jump A emb: residual	6.7e-14
ramsey jump A emb: long-run ln Y = 10%	7.4e-16
ramsey jump A emb: long-run r = closed form	4.2e-17
ramsey mixed K dis: residual	2.7e-15
ramsey mixed K dis: long-run ln Y = 10%	9.3e-16
ramsey mixed K dis: long-run r = closed form	4.2e-17
ramsey mixed K emb: residual	9.8e-12
ramsey mixed K emb: long-run ln Y = 10%	7.4e-16
ramsey mixed K emb: long-run r = closed form	4.2e-17
ramsey mixed A dis: residual	2.2e-15
ramsey mixed A dis: long-run ln Y = 10%	7.4e-16
ramsey mixed A dis: long-run r = closed form	4.2e-17
ramsey mixed A emb: residual	1.0e-12
ramsey mixed A emb: long-run ln Y = 10%	7.4e-16
ramsey mixed A emb: long-run r = closed form	4.2e-17
ramsey scurve K dis: residual	5.8e-14
ramsey scurve K dis: long-run ln Y = 10%	9.3e-16
ramsey scurve K dis: long-run r = closed form	4.2e-17
ramsey scurve K emb: residual	2.4e-15
ramsey scurve K emb: long-run ln Y = 10%	7.4e-16
ramsey scurve K emb: long-run r = closed form	4.2e-17
ramsey scurve A dis: residual	2.2e-15
ramsey scurve A dis: long-run ln Y = 10%	7.4e-16
ramsey scurve A dis: long-run r = closed form	4.2e-17
ramsey scurve A emb: residual	2.0e-15
ramsey scurve A emb: long-run ln Y = 10%	7.4e-16
ramsey scurve A emb: long-run r = closed form	4.2e-17
py jump K dis: residual	3.3e-14
py jump K dis: long-run ln Y = 10%	3.3e-16
py jump K dis: long-run r = closed form	3.3e-16
py jump K emb: residual	3.7e-15
py jump K emb: long-run ln Y = 10%	7.4e-16
py jump K emb: long-run r = closed form	1.1e-16
py jump A dis: residual	3.7e-13
py jump A dis: long-run ln Y = 10%	3.3e-16
py jump A dis: long-run r = closed form	1.1e-16
py jump A emb: residual	9.7e-14
py jump A emb: long-run ln Y = 10%	3.3e-16
py jump A emb: long-run r = closed form	1.1e-16
py mixed K dis: residual	2.1e-15
py mixed K dis: long-run ln Y = 10%	3.3e-16
py mixed K dis: long-run r = closed form	3.3e-16
py mixed K emb: residual	1.1e-11
py mixed K emb: long-run ln Y = 10%	7.4e-16
py mixed K emb: long-run r = closed form	1.1e-16
py mixed A dis: residual	2.0e-15
py mixed A dis: long-run ln Y = 10%	3.3e-16
py mixed A dis: long-run r = closed form	1.1e-16
py mixed A emb: residual	1.4e-12
py mixed A emb: long-run ln Y = 10%	3.3e-16
py mixed A emb: long-run r = closed form	1.1e-16
py scurve K dis: residual	6.3e-14
py scurve K dis: long-run ln Y = 10%	7.4e-16
py scurve K dis: long-run r = closed form	1.1e-16
py scurve K emb: residual	2.2e-15
py scurve K emb: long-run ln Y = 10%	7.4e-16
py scurve K emb: long-run r = closed form	1.1e-16
py scurve A dis: residual	2.4e-15
py scurve A dis: long-run ln Y = 10%	3.3e-16
py scurve A dis: long-run r = closed form	1.1e-16
py scurve A emb: residual	2.0e-15
py scurve A emb: long-run ln Y = 10%	3.3e-16
py scurve A emb: long-run r = closed form	1.1e-16
olg jump K dis: residual	3.6e-14
olg jump K dis: long-run ln Y = 10%	1.7e-09
olg jump K dis: long-run r = closed form	3.3e-10
olg jump K emb: residual	1.0e-13